Application of Large Language Models for Cybersecurity of Power Distribution Grids: Challenges and Opportunities

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***Abstract*—This paper explores the application of domain-specific large language models (LLMs) as a transformative approach to enhancing cybersecurity in power distribution networks. We propose a comprehensive framework that integrates LLMs with existing grid infrastructure to enable predictive analytics, proactive vulnerability assessment, intelligent threat detection, and automated incident response. By incorporating reinforcement learning (RL) agents and a human-in-the-loop mechanism, the framework supports robust and trustworthy decision-making. Leveraging LLMs' natural language processing capabilities, the system analyzes physical measurements and system logs, generates threat intelligence reports, and facilitates real-time operational decisions. The study identifies key cyber vulnerabilities in active distribution grids, including their sources, impacts, and attack taxonomy. It further discusses the challenges and risks of implementing LLMs in dynamic grid environments, alongside the opportunities they present for improving grid security. A roadmap is presented for curating domain-specific LLMs and validating their integration within standard industrial control architectures. Overall, this research demonstrates that LLMs can substantially enhance the resilience and cybersecurity of modern power distribution systems by enabling early threat detection and effective mitigation.**

***Index Terms*—Cyber-physical power system, cybersecurity, human-in-the-loop, large language model, vulnerability.**

I. Introduction

The rapid expansion of distributed energy resources (DERs) is fundamentally reshaping electric distribution grids and enhancing flexibility and sustainability, but also increasing complexity and vulnerability. As DER integration multiplies potential cyberattack vectors, traditional security mechanisms are proving insufficient against evolving threats [1]. This highlights the urgent need for advanced, real-time cybersecurity solutions that can safeguard grid reliability without hindering DER adoption [2].

Cyberattack detection in power distribution grids is typically approached through knowledge-driven or data-driven models. Knowledge-driven methods, such as rule-based detection, physical model validation, and signature analysis, rely on expert heuristics, but often fall short in managing the growing complexity of DER-rich systems. Recent developments include hybrid neural networks for IoT security [3] and edge-based solutions like Passban [4].

Data-driven techniques including machine learning, statistical anomaly detection, and deep learning, can uncover complex or previously unseen threats by analyzing large operational datasets. Notable examples include the k-smallest residual similarity (kSRS) test for false data injection attack (FDIA) detection in AC state estimation [5] and Bayesian deep learning models for handling data imbalance in intrusion detection [6]. However, these approaches often require large labeled datasets, lack adaptability to evolving threats, and provide limited interpretability.

However, both knowledge-driven and data-driven models are predominantly limited to structured numerical data from the grid's physical layer, overlooking valuable insights from unstructured sources such as system logs and operator notes. Transformer-based models, particularly large language models (LLMs), offer a compelling alternative by integrating both data types for more context-aware and effective threat detection in complex distribution grids.

LLMs have shown versatility across diverse domains, including time series analysis, operational analytics, and engineering. In the energy sector, LLMs have been applied to electricity price forecasting by incorporating market sentiment [7], and to HVAC fault diagnosis through supervised fine-tuning [8]. In cybersecurity, LLMs have outperformed conventional methods in detecting cyberattacks on smart inverters via textual command analysis [9], while other studies propose LLM-based security frameworks for IEC 61850 substations [10] and use synchronized attention mechanisms to improve log-based detection accuracy [11]. A broader survey [12] further outlines AI-driven cybersecurity approaches and associated ethical considerations.

Despite these advances, no LLM framework has been developed to address the operational and cybersecurity needs of active distribution grids. Existing studies often target generalized or device-specific scenarios, overlooking the integration challenges and unique threat landscapes associated with high-DER environments. The combined use of structured measurements and unstructured text critical for comprehensive anomaly detection remains underexplored, as do key issues such as scalability, interpretability, real-time performance, and system resilience. The key contributions of this paper are summarized as follows:

- A novel framework for integrating domain-specific LLMs with power distribution systems, demonstrating their potential for enhanced threat detection, vulnerability assessment, and real-time decision support.

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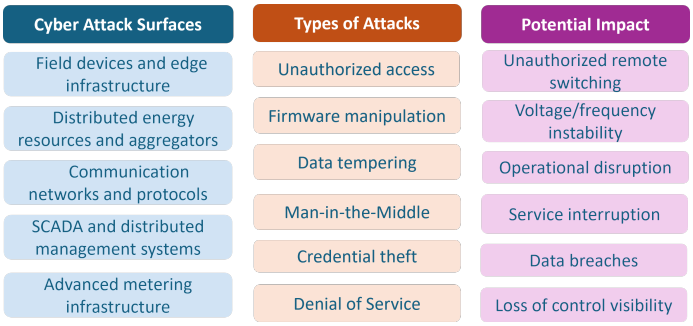


Fig. 1. Grid vulnerabilities, potential cyber attacks and consequences.

- A comprehensive review of cybersecurity vulnerabilities in active distribution grids, along with a detailed analysis of the challenges and opportunities associated with LLM implementation in critical infrastructure environments.

The remainder of the paper is organized as follows: Section II identifies cybersecurity vulnerabilities in distribution grids. Section III presents the proposed LLM framework. Section IV discusses implementation challenges, and Section V highlights the opportunities offered by domain-specific LLMs. Section VI outlines the implementation roadmap, and Section VII concludes the paper.

II. Vulnerabilities of Distribution Grids

Modern power distribution grids, while offering increased flexibility and efficiency, are becoming increasingly vulnerable to a complex array of risks [13], [14]. Understanding the sources, consequences, and classification of these vulnerabilities is essential for developing effective cybersecurity solutions in active distribution systems. Fig. 1 presents various cyber attack types, corresponding attack surfaces, and potential impacts on grid operations.

A. Sources of Vulnerabilities

The modernization of distribution grids through the integration of DERs including rooftop solar, battery storage, electric vehicles, and flexible demand-side assets has introduced significant vulnerabilities spanning physical, operational, and cyber domains. Physically, the transition to bidirectional power flows challenges legacy protection mechanisms and exposes the system to issues such as equipment overloading and voltage instability, heightened by the intermittency of renewables and reduced grid inertia from inverter-based resources. Operational vulnerabilities arise from limited real-time visibility and insufficient coordination, particularly regarding behind-the-meter DERs, complicating situational awareness and fault management. The ongoing digitalization of grid infrastructure through IoT enabled devices, cloud-based control, and increasingly open communication networks dramatically enlarges the potential cyber attack surface. Weak authentication, inadequate encryption, and fragmented ownership structures further exacerbate these risks, while regulatory and market shortcomings may allow uncoordinated or insecure DER behaviors to persist. These converging factors underscore the necessity for advanced cybersecurity frameworks capable of addressing the unique challenges of active distributed grids.

B. Consequences of Cyber Attacks

Cyber attacks targeting active distribution systems with high DER penetration can have far-reaching and disruptive consequences. Unauthorized access to control systems or devices may result in the remote manipulation or disconnection of DERs, intentional islanding, or induced voltage and frequency instabilities that threaten the reliability of the power supply. Successful attacks on critical assets or data channels can degrade power quality, cause equipment damage, or precipitate widespread outages. Moreover, data breaches, ransomware attacks, false data injections, and manipulation of system logs can obscure operational visibility, delay response efforts, and open the door to financial fraud or energy theft. In severe cases, coordinated attacks may trigger cascading failures across interconnected infrastructure, impacting not only the electricity grid but also dependent services such as telecommunications and emergency operations. The long-term effects can include diminished public trust, regulatory setbacks, and a reluctance to embrace further innovation in grid modernization.

C. Attacks Taxonomy

A clear taxonomy of cyber attacks is essential to understanding and mitigating cybersecurity risks in active distribution systems. Attacks can be categorized according to the targeted layer and the adversary's method. At the network and communication layer, attacks such as eavesdropping, denial-of-service (DoS), and man-in-the-middle (MitM) can disrupt or intercept critical data flows and control signals. Device and control layer attacks exploit vulnerabilities in smart inverters, meters, and DER controllers, enabling unauthorized switching, firmware manipulation, or forced device disconnection. Data-centric attacks including false data injection, data theft, and log tampering can undermine the integrity and availability of operational information, misleading both operators and automated control systems. Additionally, supply chain and third-party attacks may compromise software updates or cloud-based platforms, enabling persistent or large-scale disruptions. While physical attacks on field devices are less frequent, they remain a concern, especially where physical security is weak. This multifaceted taxonomy highlights the broad spectrum of cyber threats facing DER-integrated distribution systems and points to the need for layered, adaptive defense strategies, an area where large language models offer promising new capabilities.

III. Proposed LLM Cybersecurity Framework

This section introduces a framework for integrating LLMs into power distribution grids to enhance cybersecurity. The scheme leverages advanced natural language processing and reasoning capabilities of LLMs to support accurate and timely decision-making, particularly during complex or unprecedented events. The architecture of the multi-layer framework is illustrated in Fig. 2.

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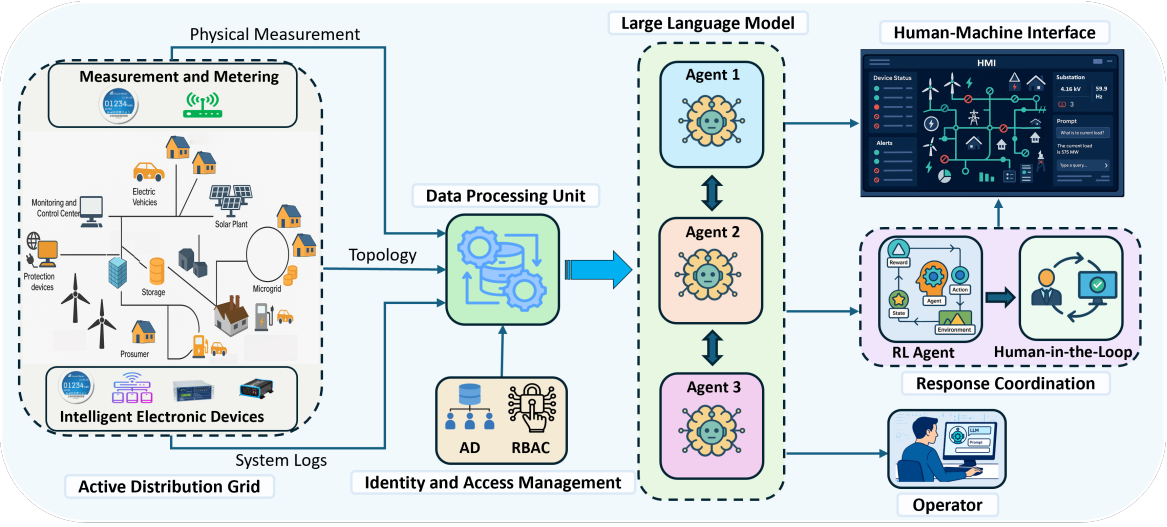


Fig. 2. Architecture of the proposed LLM cybersecurity framework.

A. LLM Integration Layer

The core of the proposed architecture is the strategic deployment of specialized LLM agents, each assigned to distinct cybersecurity functions. The framework systematically collects and processes diverse data streams to facilitate predictive security analysis, automated incident response, and reliable control actions that uphold operational standards for critical grid infrastructure.

Data Collection and Processing: This module focuses on the continuous monitoring of system elements and receiving vast amount of heterogeneous data from grid components, communication networks, and external threat intelligence sources. It maintains real-time awareness of grid configuration changes, load flow patterns, operational states, installed firmware and known security vulnerabilities. High-frequency physical measurements, such as voltages, currents, and power flow data are captured alongside system log (Syslog), which includes authentication events, configuration changes, communication errors, protocol violations, and device status updates. To address data heterogeneity, normalization processes clean and standardize both physical measurements and IT log events. This involves feature selection, removal of irrelevant Syslog and vulnerability information, and structuring datasets for efficient processing by LLM agents, while preserving the temporal alignment between physical and cyber events. This synchronization enables the detection of incidents spanning both domains, such as unauthorized access followed by anomalous grid behavior.

Domain Specific Agents: The domain-specific LLM agents receive preprocessed data from both measurement and syslog streams. They apply natural language processing techniques to interpret device communications, error messages, and operational patterns, generating human-readable threat assessments. For vulnerability analysis, the agent evaluate firmware versions, configuration settings, and communication behaviors against known vulnerability databases and threat intelligence sources [15]. It maintains device-specific security awareness and can detect weaknesses in deployed configurations or operational procedures.

A dedicated state estimation agent processes physical measurements to predict system states and identify inconsistencies indicative of false data injection attacks or compromised sensors. By understanding power flow physics, the model detects anomalies that breach standard thresholds for electrical systems. These accurate estimated states can substitute compromised data to support emergency grid operations.

An anomaly detection agent continuously analyzes both physical and syslog data, identifying subtle indicators of coordinated cyber-physical attacks. LLMs correlate seemingly unrelated events across systems and time frames, uncovering sophisticated threats that rule-based systems may overlook. They also assess user behavior, access logs, and operational activities to detect insider threats flagging deviations from normal patterns or violations of security policies. The system operates in two modes: continuous monitoring mode conducts real-time analysis across all data streams, enabling predictive threat detection through multi-agent correlation. It evaluates both the likelihood of cyber attacks and their operational impact, generating actionable control signals to isolate compromised devices and mitigate consequences. The query-based mode functions as a virtual assistant, allowing operators or security personnel to submit natural language queries for system status updates or investigative reports.

B. Response Coordination Module

To prevent disruptions caused by false alarms, all generated response actions undergo refinement stages, ensuring they do not adversely impact grid operations. This approach ensures optimal decision-making for a dynamic and critical infrastructure like power grid.

RL Agents: Response coordination is enhanced through reinforcement learning (RL) agents that improve decision-making in complex security scenarios. These pretrained agents

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adapt dynamically to system feedback, evolving threat patterns, and outcomes of past events. They continuously refine their policies to select actions that balance system reliability, security, and operational continuity. The integration of RL agents accelerates sophisticated decision-making and strengthens grid resilience against both known and emerging risks.

***Human-in-the-Loop**:* Acknowledging the importance of expert oversight in grid operations, the framework incorporates a robust human-in-the-loop mechanism. Control actions with significant operational impact such as device isolation, load shedding, or system reconfiguration are not executed automatically but instead require explicit review and approval by qualified operators. This process mitigates risks arising from model hallucinations, biases, or incomplete information, ensuring that automated recommendations do not compromise grid stability. This safeguard ensures regulatory compliance, enhances operational safety, and leverages human expertise, particularly in ambiguous or unprecedented situations.

C. Human-Machine Interface

The human-machine interface (HMI) enables comprehensive monitoring and visualization of both the power system and LLM generated cybersecurity insights. It provides a unified view of the physical grid status and cybersecurity posture through interactive dashboards, allowing personnel to explore security data, submit natural language queries to LLMs, and access detailed analysis results. Real-time alerting mechanisms prioritize notifications based on threat severity and operational impact, ensuring immediate attention to critical issues while minimizing alert fatigue from lower priority events.

IV. Challenges of Implementing LLMs

Integrating LLMs into power system cybersecurity presents challenges that extend beyond traditional IT security domains. These challenges are compounded by the criticality of power grid infrastructure, strict regulatory requirements, and the complex cyber-physical interactions inherent to distribution grids.

***Data Scarcity and Quality**:* Power distribution systems suffer from a lack of labeled datasets, limiting robust model training. Unlike general IT domains with abundant public data, grid-related information is often proprietary and classified due to security concerns [16]. Additionally, diverse communication protocols in distribution grids result in inconsistent data formats, and many field devices lack comprehensive logging, creating visibility gaps. The rarity of actual cyberattacks on power infrastructure further constrains the availability of training data for incident detection.

***LLM Deployments**:* Deploying LLMs in distribution systems is challenging due to the need for data synchronization across a geographically dispersed infrastructure comprising numerous substations, control centers, and field devices. The limited computational resources of distribution management systems (DMS) hinder on-premises deployment of LLMs, which often require substantial local processing power to meet regulatory compliance standards. Scalability is further constrained by the need to process real-time data from PMUs, SCADA systems, and security event logs.

***Adversarial Attacks and Model Security**:* Distribution networks have a broad attack surface and heterogeneous vendor ecosystems, increasing vulnerability to adversarial threats. Attackers may target LLM agents with malicious prompts, compromise edge device inputs, or exploit weaknesses in data aggregation, thereby manipulating detection outcomes or concealing threats. Ensuring model integrity in this distributed, dynamic environment remains a significant challenge.

***Model Hallucination and Biases**:* LLMs may generate fabricated threats or non-existent vulnerabilities, leading to false alarms, inappropriate switching actions, and misallocation of field resources. Such hallucinations can result in operational disruptions, especially during time-critical incidents. Furthermore, biases in training data regarding threat origins, geographic vulnerability, or equipment reliability can cause unequal threat assessment and blind spots in security monitoring.

***Trust, Confidentiality and Regulatory Compliance**:* Power system operators require explainable and reliable security insights to support informed decision-making. False positives suggested by LLM agent can lead to unnecessary equipment isolation or load shedding, potentially compromising grid stability. LLMs that process operational, customer, or system data must adhere to strict confidentiality, access control, and auditability standards. Compliance with regulations such as NERC CIP and local data privacy laws demands rigorous oversight of data storage, transmission, and processing.

***Cost and Resource Allocation**:* Many distribution utilities, particularly smaller municipal or rural ones, operate under constrained budgets and workforce limitations. The high computational and storage demands of LLMs, especially for on-premise deployment required by regulatory frameworks, can be cost-prohibitive. Without shared-service models or consortium-based approaches, widespread adoption may remain out of reach for resource-limited utilities.

V. Opportunities with Domain-Specific LLMs

Despite existing challenges, LLMs offer transformative opportunities to enhance cybersecurity and operational resilience in power distribution systems. By leveraging advanced pattern recognition, contextual reasoning, and automation capabilities, these models can augment traditional tools and address limitations in current practices.

***Proactive Vulnerability Assessment**:* Domain-specific LLM agents can improve vulnerability assessment by continuously analyzing device configurations, network topologies, and operational patterns to identify risks before exploitation occurs. These agents understand the interdependencies between electrical and cyber components, maintain updated security statuses for field devices and communication links, and synthesize information from equipment manuals and firmware advisories. As a result, they can uncover non-obvious attack vectors often missed by conventional tools and recommend timely mitigation strategies.

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Predictive Anomaly Detection: LLMs are well-suited for predictive anomaly detection due to their ability to learn complex spatio-temporal patterns. By integrating real-time physical measurements, such as voltage, current, or power fluctuations with cyber event logs, LLMs can identify subtle anomalies indicative of compromised devices or evolving threats. This capability is especially effective in detecting advanced persistent threats that may remain dormant before activation. The predictive approach enables early warning, focused investigation, and reduced risk to grid stability.

Incident Response Optimization: Domain-specific LLM agents support rapid and effective incident response while maintaining operational continuity and safety. By mapping detected threats to predefined response playbooks and offering context-aware recommendations, LLMs assist operators in prioritizing mitigation efforts and resource allocation. This intelligent guidance shortens response times, limits disruption, and ensures protection of critical assets in line with operational protocols.

Log Analysis and Correlation: Unlike conventional machine learning models focused solely on physical data, LLMs excel in parsing, normalizing, and correlating vast volumes of cybersecurity and operational logs from SCADA systems, protective relays, communication networks, and security devices. They reveal relationships between seemingly isolated events, enabling detection of coordinated attacks and persistent threats. This comprehensive visibility is difficult to achieve through manual analysis alone.

Security Documentation and Reporting: LLMs can automate the generation of incident reports, compliance documents, and audit trails by extracting relevant details from logs and operator notes. These models understand domain-specific terminology and regulatory formatting requirements, enabling consistent and accurate reporting aligned with standards such as NERC CIP. This automation reduces administrative workload, improves documentation quality, and streamlines compliance processes.

Automated Penetration Testing: Domain-specific LLMs can simulate tailored penetration testing scenarios by combining knowledge of cyber attack techniques with power system operational constraints. They can probe for weaknesses in configurations and network segmentation, assess system responses, and support ongoing security evaluations without costly manual testing. This enables utilities to validate defenses, adapt to emerging threats, and maintain a strong security posture.

VI. Roadmap for Implementation

A. Compatibility with Existing System

To enable secure and scalable deployment of the proposed LLM-based cybersecurity framework within real-world power distribution environments, it is critical to ensure compatibility with established industrial models, such as the Purdue reference architecture for industrial control systems. The modular structure of the framework is specifically designed to align with the multilayered Purdue model, facilitating seamless integration into existing grid infrastructure. During deployment, domain-specific LLM agents are embedded into the operational environment and interfaced with the grid's data collection, analytics, and control systems. Fig. 3 illustrates how the framework maps onto the Purdue model.

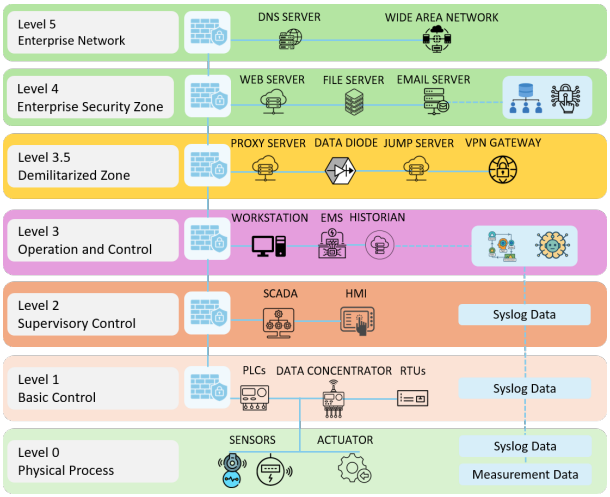


Fig. 3. Proposed LLM framework deployment in Purdue ICS model.

Physical power system elements such as sensors, actuators, relays, circuit breakers, meters, programmable logic controllers (PLCs), and remote terminal units (RTUs) at Levels 0 and 1 of the Purdue model provide both real-time and historical data in the form of physical measurements and system logs, serving as essential inputs for LLM training and operation. Level 2 houses the SCADA system and the human-machine interface, which ensure the timely transmission of distribution grid data to the LLM agents. Together, the elements from Levels 0 to 2 form the foundational data infrastructure supporting the LLM framework.

At Level 3, the operator's workstation, energy management system (EMS), and data historian closely interact with the LLMs. This layer integrates RL agents and HMI capabilities, supporting adaptive decision-making while complying with NERC CIP cybersecurity standards. All agents and decision-making modules operate within the trusted network below the demilitarized zone (DMZ), ensuring they reside in the most secure operational tier. The active directory (AD) and role-based access control (RBAC) are implemented at Level 4 Enterprise Network for user authentication and access management. While essential for enforcing security policies across the organization, these systems do not directly influence the decision-making processes of the LLM agents.

B. Domain-Specific LLM Curation

The adaptation of pretrained LLMs for power system cybersecurity requires fine-tuning with domain-specific data such as electrical measurements, device configurations, logs, and incident records to capture normal and anomalous grid behaviors. This enables the models to detect subtle cyber threats in context and operate effectively in both continuous monitoring and prompt-based support modes.

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TABLE I
PRETRAINED LLMS FOR POWER GRID CYBERSECURITY APPLICATIONS

Model	Cybersecurity Functionality	Limitations/ Dependencies	On-Premises
BERT/ RoBERTa/ Distil-BERT	SCADA log parsing, incident classification, anomaly detection, event correlation	Requires domain-specific fine-tuning; limited real-time processing capabilities	Yes
CyberBERT/ SecBERT	ICS malware detection, substation event classification, OT network threat hunting	Focused on logs, not physical data; requires labeled datasets for domain adaptation	Yes
CyberGPT/ ELECTRA	Real-time data monitoring, threat detection, cross-domain attack correlation	Needs cybersecurity fine-tuning; limited interpretability for regulatory use	Yes
LLaMA 2/ LLaMA 3	Customizable threat analysis, privacy-sensitive grid operations	Requires extensive fine-tuning; high GPU demand for deployment	Yes
GPT-3/ GPT-4	Advanced threat intelligence, incident response, compliance guidance	API-only access; data privacy concerns; latency limits real-time use	No
Claude	Security planning, safety-focused threat assessment	Proprietary cloud service; limited customization options	No
Gemini	Multimodal threat analysis, network topology visualization	Google Cloud dependency; limited validation for utility use	No

Several types of pretrained LLMs offer promising capabilities for power grid cybersecurity application. Transformer-based models like BERT are effective for log analysis and anomaly detection due to their ability to capture contextual dependencies in data. GPT-family models (e.g., GPT-3, GPT-4) are particularly valuable for natural language processing tasks such as generating incident summaries, synthesizing threat intelligence, and supporting operator interactions. LLaMA models provide customizable, open-source alternatives suitable for on-premises deployment, which is often required for regulatory compliance. Models like Google's Gemini introduce multimodal capabilities that correlate text-based events with visual data, while Anthropic's Claude models emphasize safety and reduced hallucination, supporting accuracy-critical applications such as compliance documentation. Additionally, cybersecurity-specialized models like CyberBERT, SecBERT, and CyberGPT offer enhanced domain knowledge for threat identification and response.

While many of these models show promise for cybersecurity applications, open-source, on-premise deployable options are particularly well-suited due to regulatory requirements and the need for greater control over data privacy, security, and infrastructure integration. Table I summarizes the functionalities, limitations, and deployment considerations of various LLMs in the context of power grid cybersecurity.

VII. CONCLUSIONS

This research demonstrates that domain-specific LLMs present a significant opportunity to address the increasing complexity and sophistication of cyber threats targeting power distribution grids. A comprehensive analysis of cyber vulnerabilities in active distribution networks highlights the limitations of traditional cybersecurity approaches in managing the dynamic challenges introduced by smart grid technologies.

The proposed framework shows that integrating LLM agents with power distribution infrastructure offers a systematic means of leveraging AI-driven capabilities while upholding the reliability and security standards essential for critical infrastructure. Despite challenges related to computational demands, model reliability, cost, and regulatory compliance, the potential benefits of domain-specific LLMs outweigh these limitations. As cyber adversaries continue to advance in scale and complexity, implementing the proposed framework will meaningfully strengthen the cybersecurity posture of power distribution systems, contributing to a more resilient and secure energy infrastructure.

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